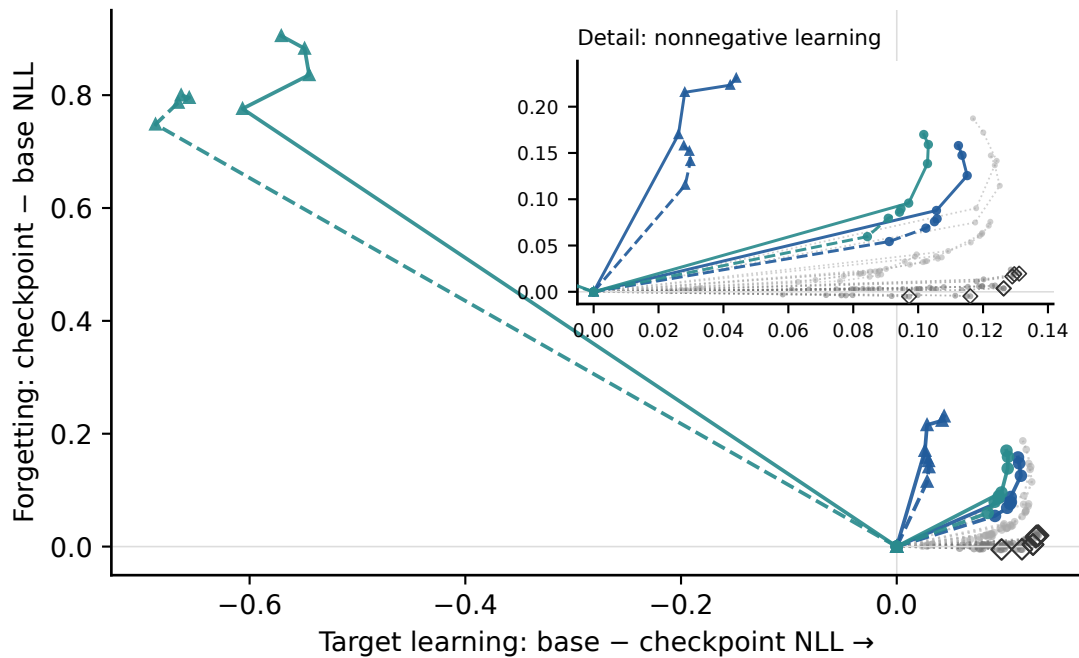
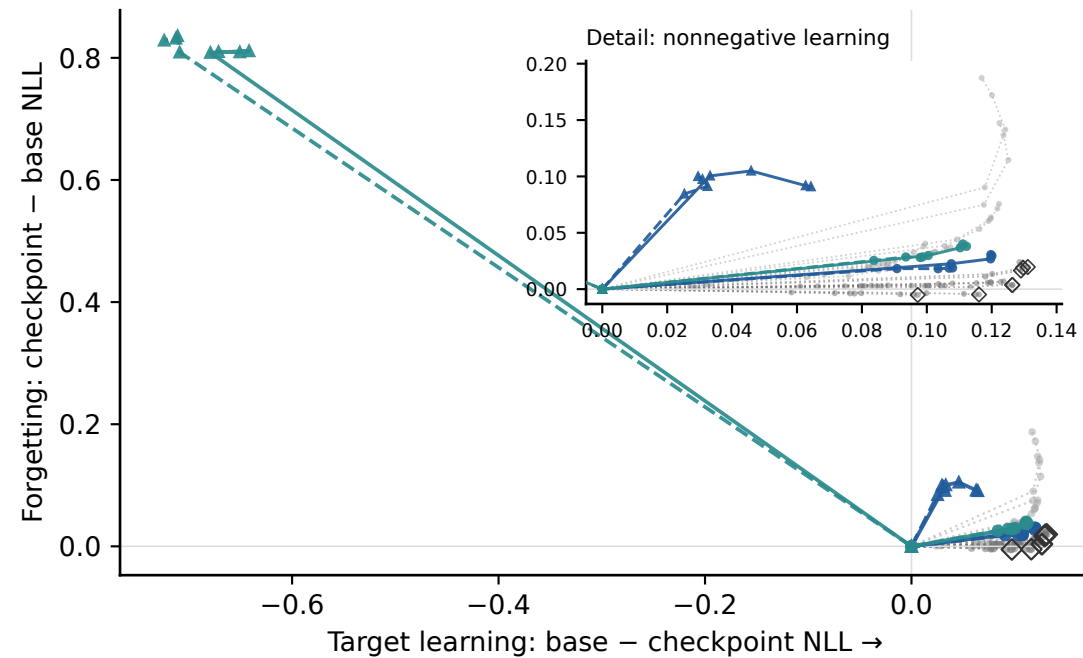


Training-time adaptation and feedback for continual pretraining

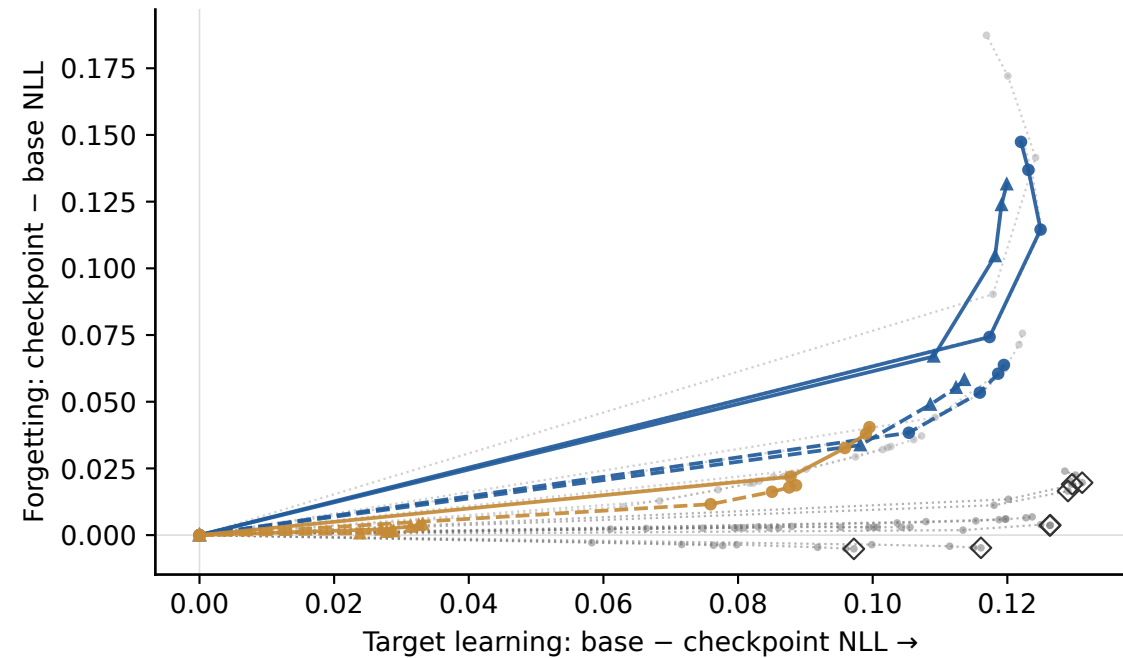
a Target-only feedback



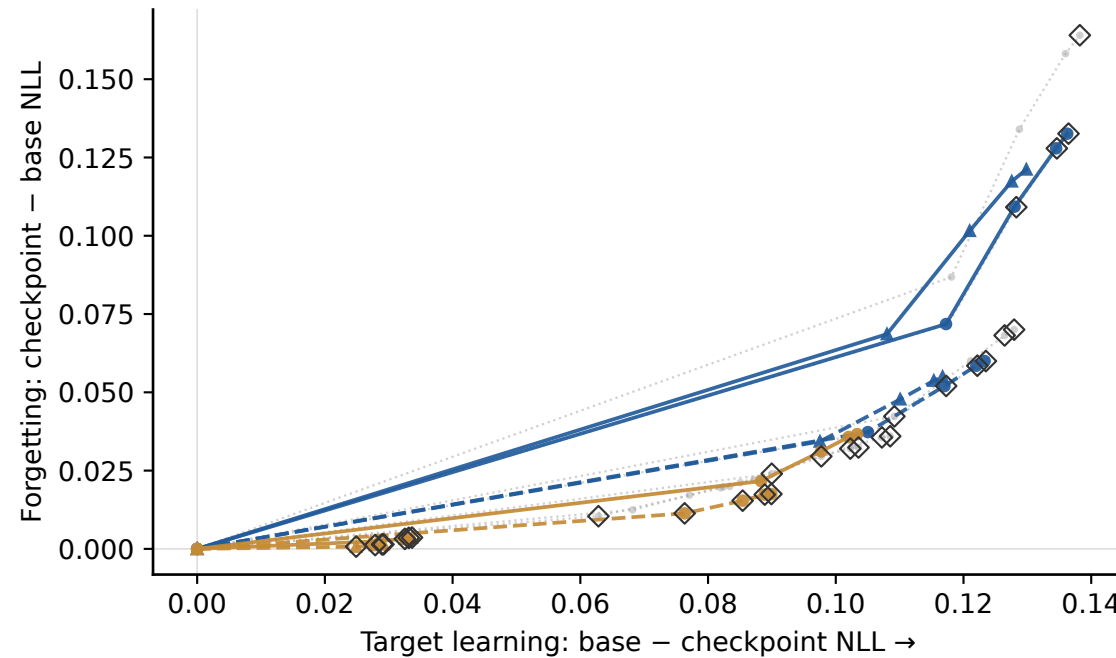
b Retention-aware feedback



c Teacher feedback · quarter data



d Teacher feedback · one pass



— First-order feedback — Frozen-initial teacher ◇ Observed ordinary frontier
— Exact / adapted teacher ⋯ Ordinary controls

All 60 trajectories: 52 new and 8 reused ordinary controls; four fixed nonzero checkpoints each. Lines connect measured checkpoints only.
 More learning is rightward; less forgetting is downward. Ordinary controls include replay and fixed-base distillation; frontier markers are measured points.
 Treatment dashed/solid: outer LR $1e-5$ / $3e-5$. Circle/triangle: ρ 0.03 / 0.3 above, KL beta 1 / 10 below.
 Top insets magnify the nonnegative-learning region; the main panels retain every point and the full observed ranges.
 Observed 240/240 points; 0 missing points are gaps and remain listed in the exported data.
 SmoLM2-135M, math CPT, one seed, examined 512-document/domain holdouts. Quarter and one-pass regimes have different unique-data budgets.